Mixed-Integer Linear Programming

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EgyE 332

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Outline

1. Motivational Example
2. Branch-and-Bound
3. Examples in Energy Engineering
    a. Simple Unit Commitment
    b. Unit Commitment with Renewables and Storage
    c. Waste-to-Energy Facility Location-Allocation

Mixed-Integer Linear Programming

Energy Investment Portfolio Selection

https://github.com/kspilario/EgyE332_opti/blob/main/MILP%20Problems/energy_investment_knapsack.ipynb

An energy company has a limited capital budget and must decide which candidate energy projects to invest in.

Each project has:

- a fixed investment cost, and
- an expected annual energy production.

A project is either **selected completely** or **not selected at all**. Partial investment is not allowed.

The objective is to select the combination of projects that **maximizes total annual energy production** without exceeding the available investment budget.

Project	Technology	Investment Cost (₱ million)	Annual Energy Production (GWh/year)
1	Solar Farm	40	55
2	Wind Farm	50	75
3	Small Hydro	60	85
4	Biomass Plant	35	45
5	Rooftop Solar Program	25	30

Budget: PhP 100 million

$$\max \quad 55x_1 + 75x_2 + 85x_3 + 45x_4 + 30x_5$$

$$s.t. \quad 40x_1 + 50x_2 + 60x_3 + 35x_4 + 25x_5 \leq 100$$

$$x_i \in \{0, 1\}$$

This is an 0-1 Knapsack Problem.

Mathematical Programs

In general, the method for solving mathematical models is highly dependent on the **type of model**:

TYPE	Objective Function	Constraints	Decision Variables
Linear Program (**LP**)	Linear	Linear	Continuous
Nonlinear Program (**NLP**)	At least the objective function or 1 constraint is nonlinear		Continuous
Integer Program (**IP**)	Linear or nonlinear	Linear or nonlinear	All Integer
Mixed-integer Linear Program (**MILP**)	Linear	Linear	Some Integer; Others Continuous
Mixed-integer Nonlinear Program (**MINLP**)	At least the objective function or 1 constraint is nonlinear		Some Integer; Others Continuous

We'll discuss **Branch-and-Bound Method.**

Source: Chap.14; El-Halwagi, M., 2012. *Sustainable Design Through Process Integration.* USA: Elsevier Inc.

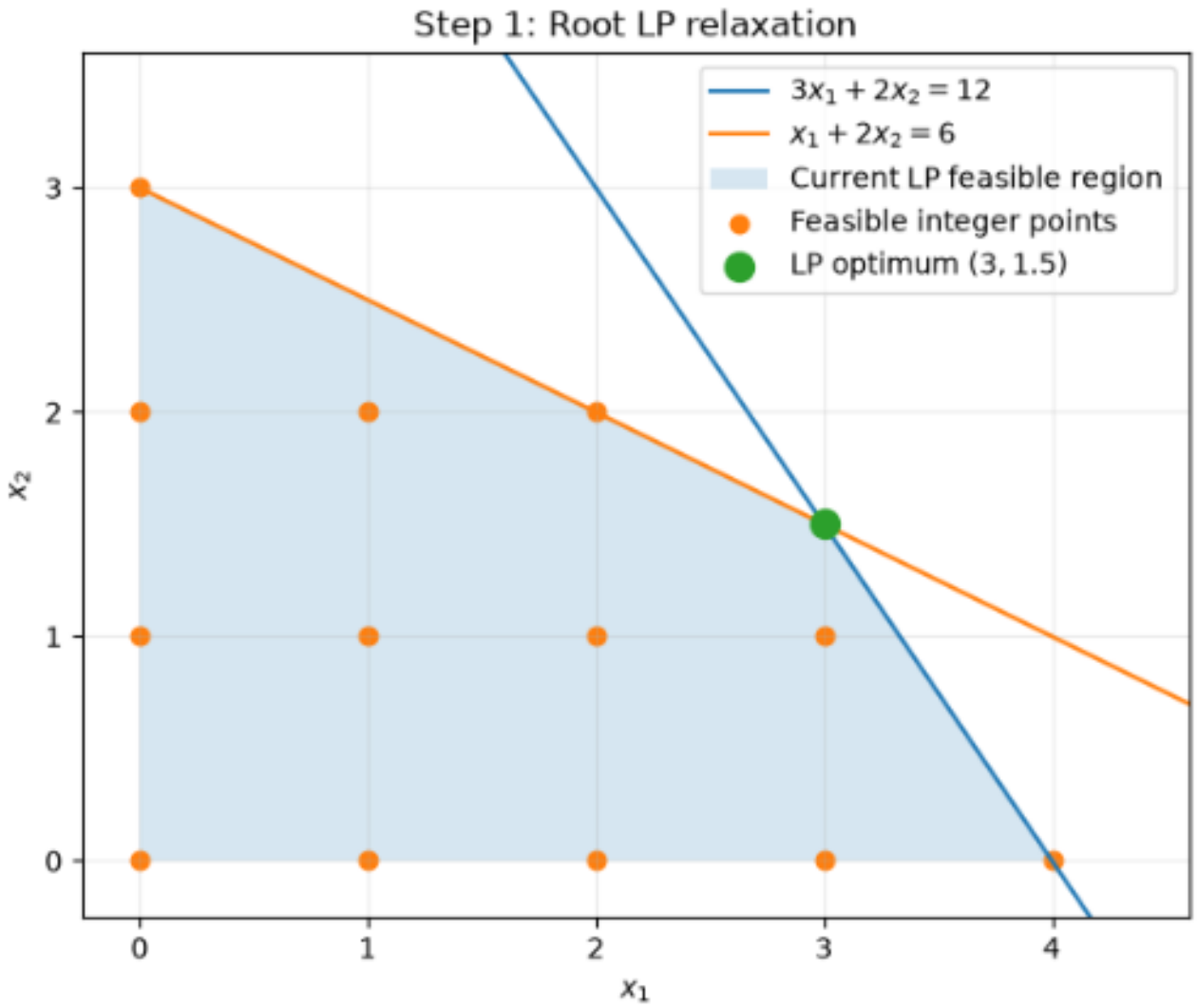
*The word "program" is due to George Dantzig (1974).

Recall: Linear Programming

Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ s.t. \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

But for now, what if we let x_1, x_2 be continuous and solve the **LP** first?

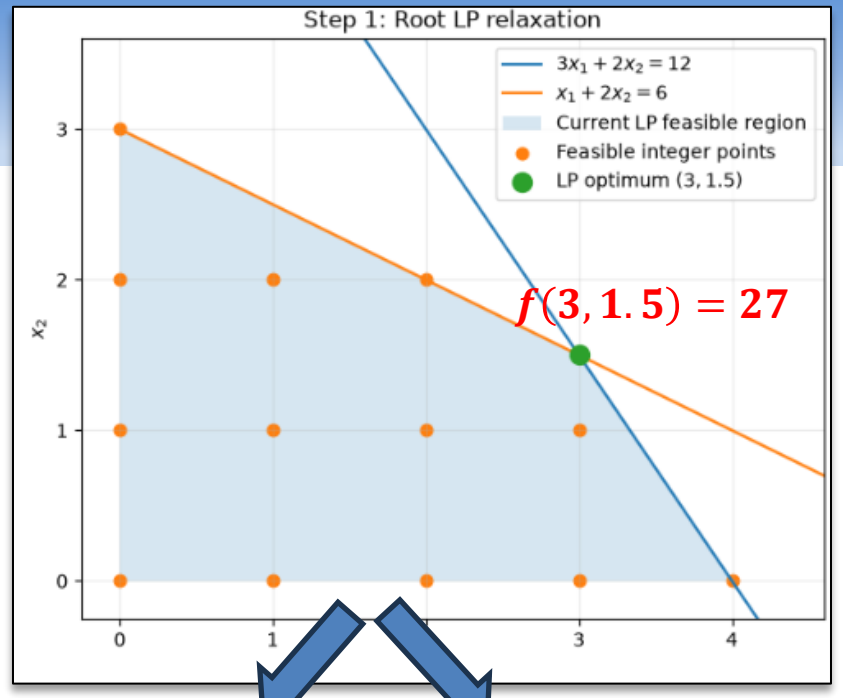


Branch-and-Bound Method

Consider an **integer program**:

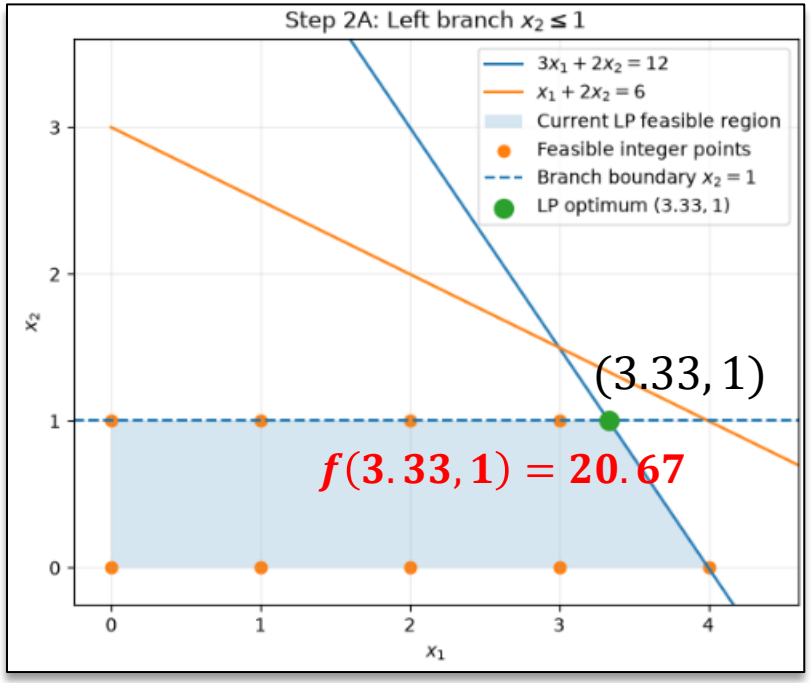
$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ s.t. \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

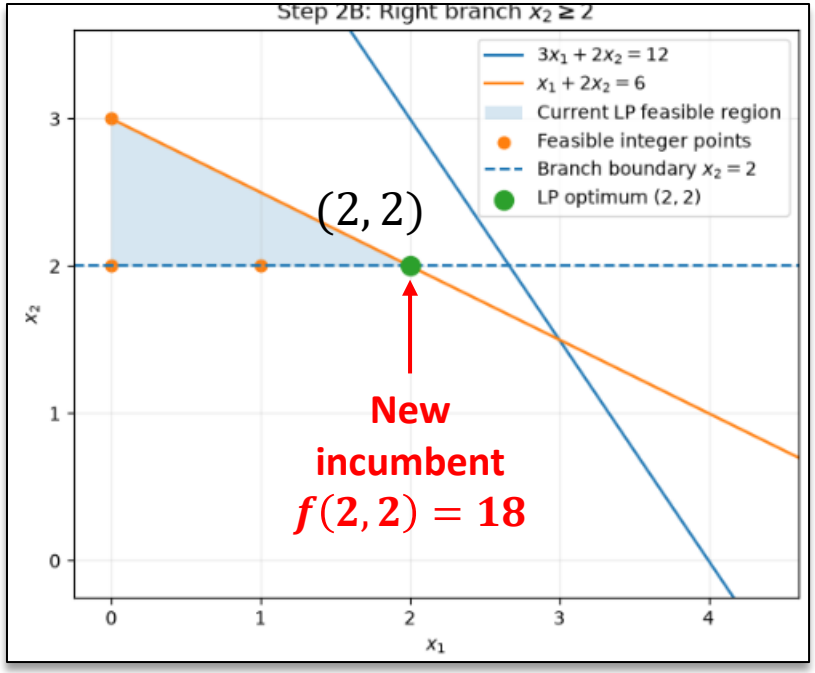
In branching, we split the non-integer solution into two, then solve the LP in each.



$x_2 \leq 1$

$x_2 \geq 2$

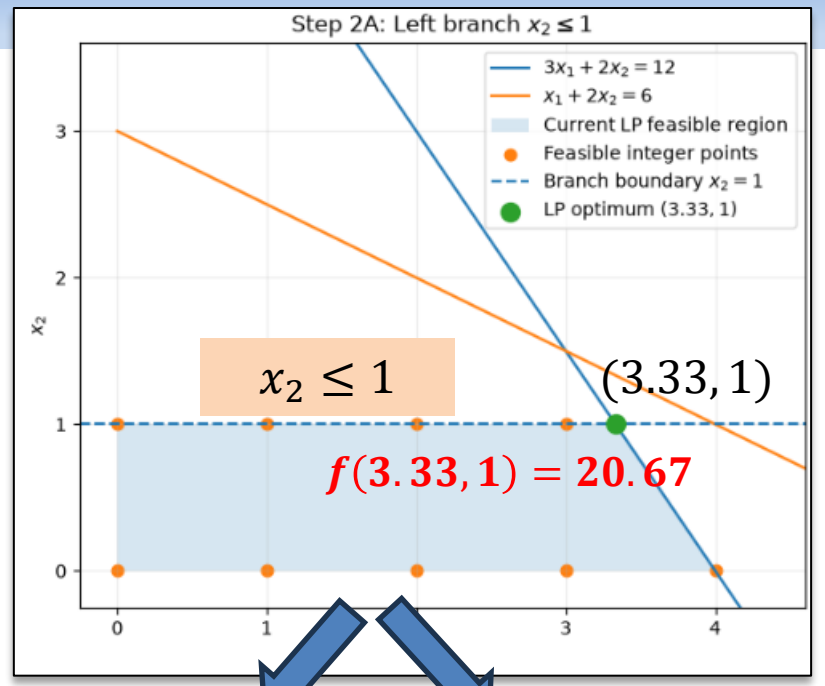


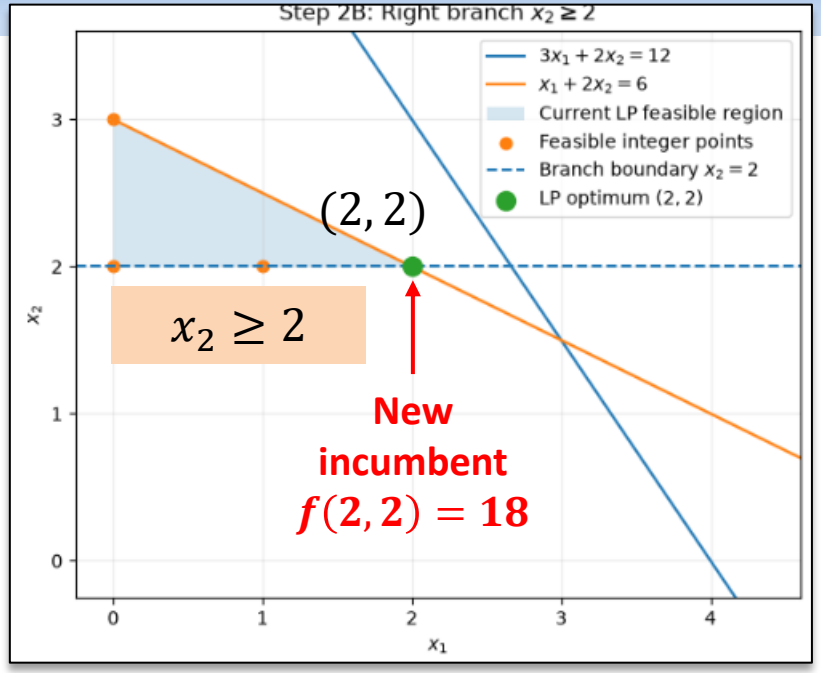


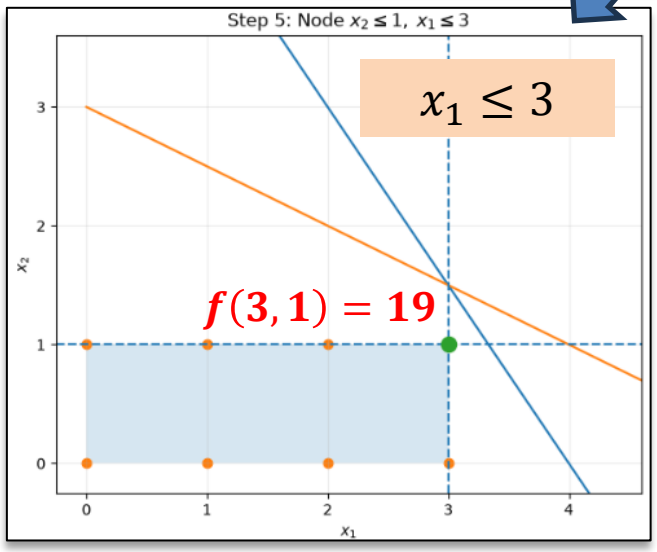
Branch-and-Bound Method

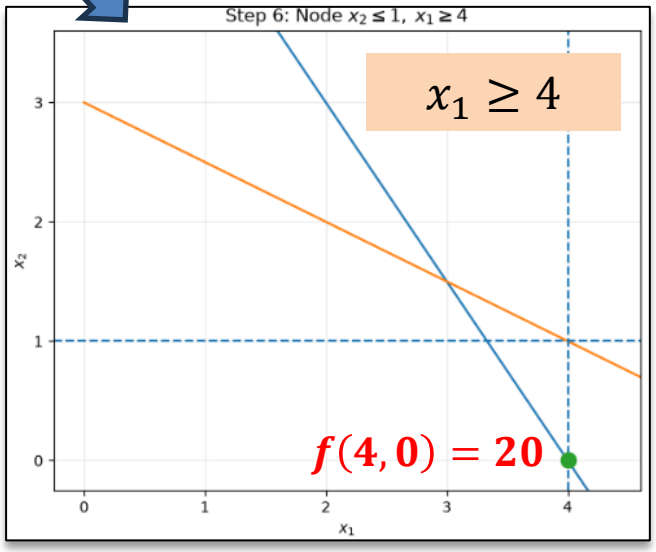
Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ s.t. \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$









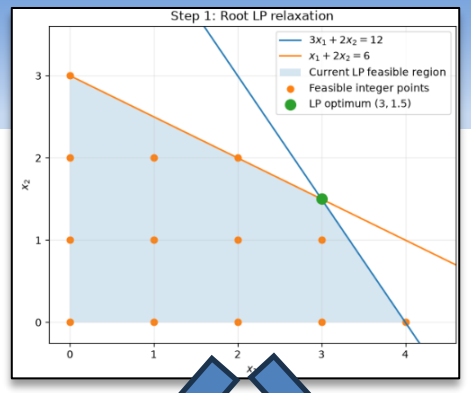
We branch on the left and right of $(3.33, 1)$ since it still has a fractional part.

Branch-and-Bound Method

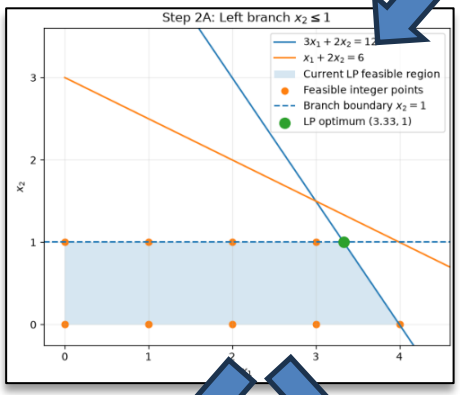
Consider an **integer program**:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ s.t. \quad & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 2x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

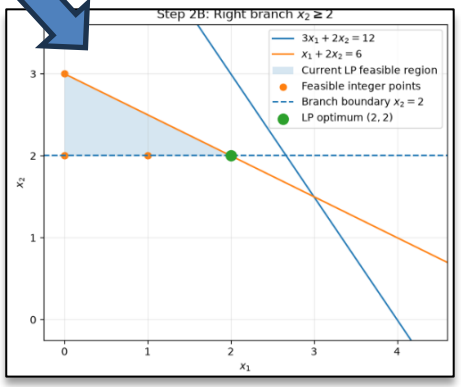
In summary:



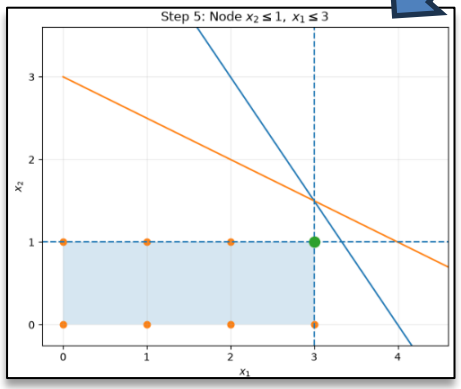
$f(3, 1.5) = 27$



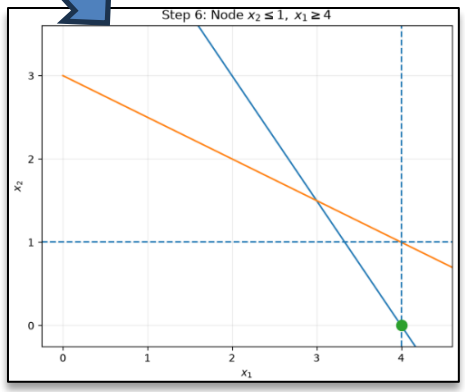
$f(3.33, 1) = 20.67$



$f(2, 2) = 18$



$f(3, 1) = 19$



$f(4, 0) = 20$

We stop because there are no unresolved (active) nodes left.

All solutions are already the best in each region.

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    a. **Simple Unit Commitment**
    b. **Unit Commitment with Renewables and Storage**
    c. **Waste-to-Energy Facility Location-Allocation**

Unit Commitment: Simple

A very simple multi-period unit commitment problem is shown below:

$$\min_{P_{g,t},u_{g,t},y_{g,t}} \quad \sum_t \sum_g c_g^{var} P_{g,t} + c_g^{NL} u_{g,t} + c_g^{SU} y_{g,t}$$

$$s.t. \quad \sum_g P_{g,t} = D_t$$

$$P_g^{min} u_{g,t} \leq P_{g,t} \leq P_g^{max} u_{g,t}$$

$$u_{g,t} - u_{g,t-1} = y_{g,t} - z_{g,t}$$

$$y_{g,t} + z_{g,t} \leq 1$$

$$P_{g,t} - P_{g,t-1} \leq R_g^{up} + P_g^{min} y_{g,t}$$

$$P_{g,t-1} - P_{g,t} \leq R_g^{down} + P_g^{min} z_{g,t}$$

*Assume a total period of 15 hrs and 1 period = 1 hr.

Decision Variables:

$P_{g,t}$ = power output of generator g at time t

$u_{g,t} \in \{0,1\}$ = indicates if gen g is online at time t

$y_{g,t} \in \{0,1\}$ = indicates if gen g undergoes startup at time t

$z_{g,t} \in \{0,1\}$ = indicates if gen g undergoes shutdown at time t

Other parameters:

c_g^{var} = variable generation cost of generator g

c_g^{NL} = no-load cost of generator g

c_g^{SU} = startup cost of generator g

P_g^{min}, P_g^{max} = min and max power output for generator g

R_g^{up}, R_g^{down} = ramp-up and ramp-down constraint for generator g

D_t = demand at time t

Unit Commitment with Renewables and Storage

A 24-hour unit commitment problem is shown below:

$$\min_{P_{g,t},u_{g,t},y_{g,t}} \quad \sum_t \left[\sum_g \left(c_g^{var} P_{g,t} + c_g^{SU} y_{g,t} \right) + c_B \left(P_{ch,t} + P_{dis}, t \right) \right]$$

$$s.t. \quad P_{s,t} + P_{w,t} + P_{dis,t} + \sum_g P_{g,t} = D_t + P_{ch,t}$$

$$P_g^{min} u_{g,t} \leq P_{g,t} \leq P_g^{max} u_{g,t}$$

$$u_{g,t} - u_{g,t-1} = v_{g,t} - w_{g,t}$$

$$v_{g,t} + w_{g,t} \leq 1$$

$$P_{g,t} - P_{g,t-1} \leq R_g^{up} + P_g^{min} v_{g,t}$$

$$P_{g,t-1} - P_{g,t} \leq R_g^{down} + P_g^{min} w_{g,t}$$

$$0 \leq P_{s,t} \leq S_t^{avail}$$

$$0 \leq P_{w,t} \leq W_t^{avail}$$

$$E_t = E_{t-1} + \eta_{ch} P_{ch,t} - \frac{P_{dis,t}}{\eta_{dis}}$$

$$P_{ch,t} \leq P_{ch}^{max} z_t$$

$$P_{dis,t} \leq P_{dis}^{max} (1 - z_t)$$

$$0 \leq E_t \leq E^{max}$$

$$E_{24} = E^{init}$$

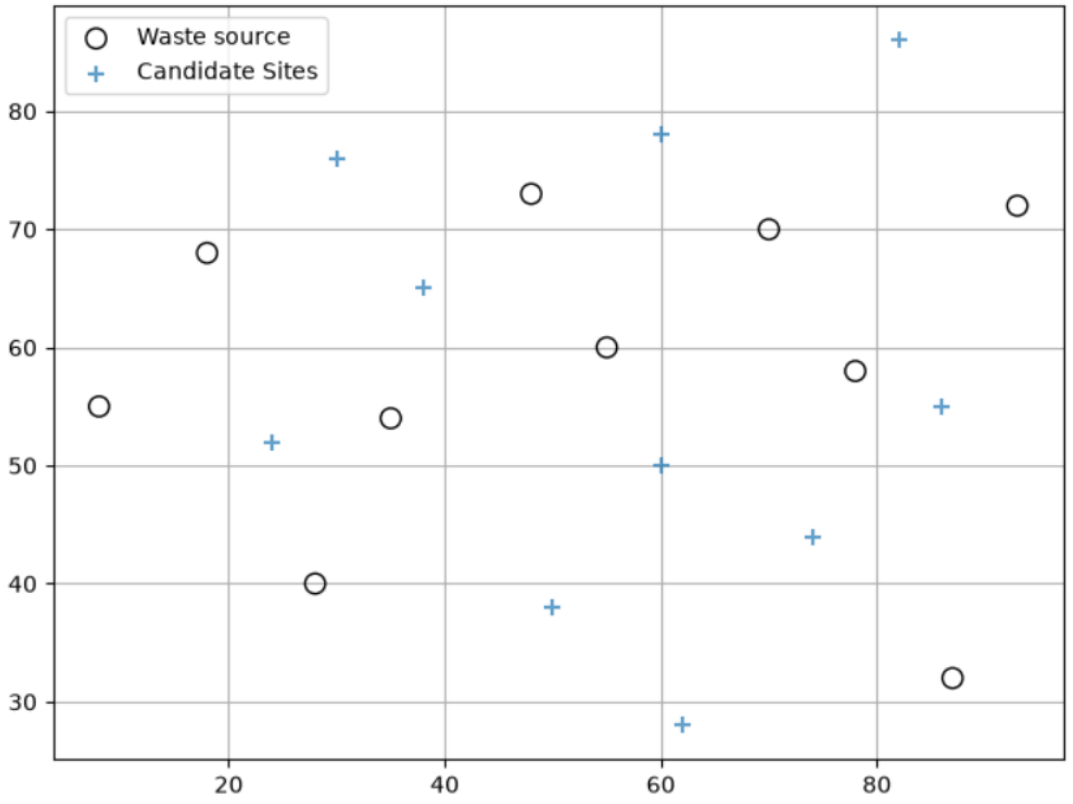
* Assume a total period of 24 hrs and 1 period = 1 hr.

Waste-to-Energy Facility Location-Allocation Planning

Our goal is to find the best location, best technology, and best transport of wastes in the following map:

$$\min_{z_{s,k},a_{i,s},Q_{s,k}} \quad CRF \sum_s \sum_k K_k z_{s,k} + \sum_s \sum_k c_k^{op} Q_{s,k} + c^{tr} \sum_i \sum_s W_i d_{i,s} a_{i,s} - p^E \sum_s \sum_k e_k Q_{s,k}$$

k	Technology
1	Pyrolysis
2	Incineration
3	Anaerobic Digestion



Decision Variables:

$z_{s,k} \in \{0,1\}$ = indicates if tech k is built at site s

$a_{i,s} \in \{0,1\}$ = indicates if waste source i is assigned to site s

$Q_{s,k}$ = annual waste processed at site s using tech k [ton/yr]

Fixed parameters:

CRF = capital recovery factor given discount rate r and 10-yr project life

K_k = upfront installation cost for tech k [$]

c_k^{op} = operating cost for tech k [$/ton]

c^{tr} = waste transportation cost [$/ton-km]

e_k = net saleable energy yield [MWh/ton]

p^E = energy selling price [$/MWh]

$d_{i,s}$ = straight-line distance from source i to site s

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